Introduction

When many hypothesis tests are performed simultaneously, the probability that at least one rejects a true null grows rapidly. The **Bonferroni correction** is the simplest, most conservative way to control the family-wise error rate (FWER): divide the target α by the number of tests.

Prerequisites

Hypothesis testing, Type I error.

Theory

For m tests each at nominal α, if all nulls are true, the probability of at least one false rejection is

$$P(\text{any rejection}) \le m\alpha$$

by the union bound. To keep the overall rate below α, test each hypothesis at $\alpha_{\text{per}} = \alpha/m$.

Equivalently, multiply each raw p-value by m and compare to α:

$$p_j^{\text{adj}} = \min(1, m p_j).$$

Properties:

- Always controls FWER at exactly α regardless of dependence structure among tests.
- Very conservative when tests are positively correlated.
- Powerful only when m is small; the correction becomes harsh as m grows.

Assumptions

No distributional assumptions about the tests themselves; Bonferroni works for arbitrary dependence. The cost is a lower power for each individual test.

R Implementation

```r
set.seed(2026)

# Ten simulated test statistics under a mix of null and alternative
n_each <- 30
groups <- LETTERS[1:11]
y <- c(rnorm(n_each, 50, 10),        # group A (control)
       rnorm(n_each, 50, 10),        # B null
       rnorm(n_each, 52, 10),        # C small effect
       rnorm(n_each, 48, 10),        # D
       rnorm(n_each, 55, 10),        # E real effect
       rnorm(n_each, 50, 10),        # F null
       rnorm(n_each, 58, 10),        # G large real effect
       rnorm(n_each, 51, 10),        # H
       rnorm(n_each, 49, 10),        # I
       rnorm(n_each, 53, 10),        # J
       rnorm(n_each, 50, 10))        # K
grp <- rep(groups, each = n_each)

# Pairwise t-tests vs group A
```

```
p_raw <- sapply(groups[-1], function(g)
  t.test(y[grp == g], y[grp == "A"])$p.value)

p_bonf <- p.adjust(p_raw, method = "bonferroni")
p_holm <- p.adjust(p_raw, method = "holm")

data.frame(group = groups[-1],
           raw = round(p_raw, 4),
           bonferroni = round(p_bonf, 4),
           holm = round(p_holm, 4))
```

Output & Results

```
  group     raw  bonferroni   holm
1     B  0.5734      1.0000 1.0000
2     C  0.1832      1.0000 0.9161
3     D  0.9112      1.0000 1.0000
4     E  0.0188      0.1880 0.1506
5     F  0.4234      1.0000 1.0000
6     G  0.00041     0.0041 0.0041
7     H  0.8344      1.0000 1.0000
8     I  0.7122      1.0000 1.0000
9     J  0.1643      1.0000 0.9161
10    K  0.4721      1.0000 1.0000
```

Only the large-effect test (G) survives Bonferroni correction at $\alpha = 0.05$. Several raw p-values below 0.05 (e.g., E at 0.019) become non-significant after correction.

Interpretation

"After Bonferroni correction for 10 pairwise comparisons (adjusted $\alpha = 0.005$), only group G differed significantly from control (raw p = 0.0004, adjusted p = 0.004)."

Practical Tips

- Bonferroni is overly conservative when m is large; prefer Holm or FDR.
- `p.adjust()` in R supports "bonferroni", "holm", "hochberg", "hommel", "BH" (FDR), "BY" methods.
- Always apply the correction to a well-defined family of tests pre-specified in the protocol.
- Do not apply Bonferroni to a handful of primary pre-specified comparisons when they were individually justified.
- For a single primary endpoint and secondary exploratory analyses, correction often applies only to the exploratory family.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests